

# Precise Determination of the Restgas Profile for High-Precision Luminosity Determination

**Jinxin Li**

Ruhr-university Bochum | Experimental physics I

March 30, 2026

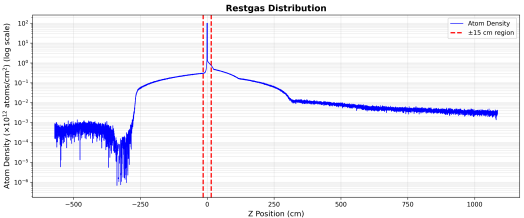
# Restgas Distribution & Luminosity Determination

The Restgas Profile

- Simulated from Münster.
- Continuous distribution along the beam pipe.

Previous Achievements

- `LuminosityFit` utilizes Restgas for luminosity determination.
- Proven precision:
  $\Delta\mathcal{L}/\mathcal{L} < 0.5\%$
  (for $p_{lab} = 1.5 - 15$ GeV/$c$)

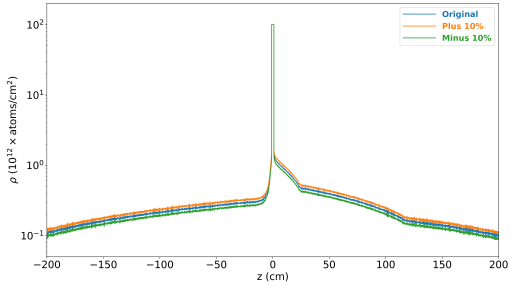



*Restgas density distribution along the $z$-axis*

# Reality in the Experiment

Simulated Restgas profiles may differ from the real distribution.

**Sensitivity Study:**




**Fit Model**

Original Simulated Restgas profile.

**Real data**

Original or $\pm 10\%$ Restgas profiles.

# Reality in the Experiment

Simulated Restgas profiles may differ from the real distribution.

**Sensitivity Study:**

| Target Profiles | $\Delta\mathcal{L}/\mathcal{L}$ with Consistent Model | $\Delta\mathcal{L}/\mathcal{L}$ with Original profile |
|---|---|---|
| -10% Restgas | $-0.02 \pm 0.08$ | $-3.39 \pm 0.08$ |
| Original | $0.02 \pm 0.08$ | $0.02 \pm 0.08$ |
| +10% Restgas | $0.02 \pm 0.08$ | $3.39 \pm 0.08$ |

High precision in luminosity determination requires a precise Restgas profile.

# Reconstruction Bottleneck

**Why standard reconstruction fails for Restgas background?**

- Beam-Restgas interactions distributed along the entire beam pipe.
- Standard Track finder (SttMvdGem/Barrel) optimized for IP at $(0, 0, 0)$.
- PID step back-propagated tracks to the $z$-axis for vertex determination.

## Consequences of Displaced Tracks

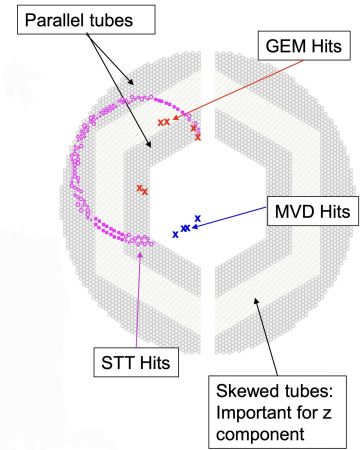
- Wrong reconstructed vertex.
- Bias in $dE/dx$ and momentum vector measurements.
- Low reconstruction efficiency.

# Track Finder for Displaced Vertices

## Apollonius Triplet Track Finder - A Tailored Track Finder for Secondaries

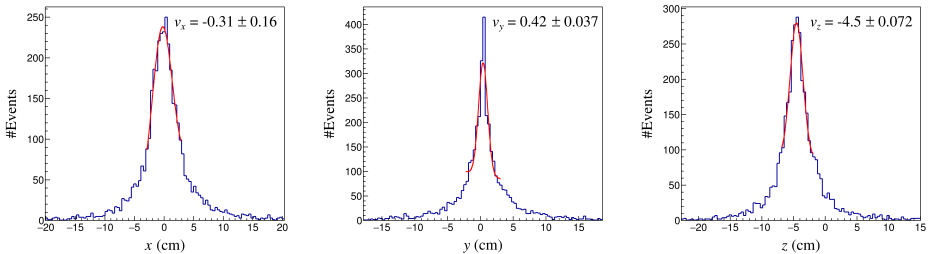
1. Apollonius circles in $xy$-plane
   - Select STT hits
   - Calculate circles
   - Chose a proper subset circle
2. Longitudinal reconstruction in $z$-axis
   - Pz Finder with skewed STT hits
   - Calculate $z$ position and $p_z$
3. Global Tracking
   - KalmanFilter with MVD & GEM hits.

*Developed by Anna Alicke, JÜLICH*



# A POCA-First Approach:

1. PID step without back-propagation.
2. Vertex determination via POCA of all $\bar{p}p$ tracks.




*simulated events with a true vertex at* $x = -0.2\,\mathrm{cm},\ y = 0.4\,\mathrm{cm},\ z = -5.0\,\mathrm{cm}$

3. PID step with back-propagation to the POCA vertex (*POCA-to-point*).

# Performance of POCA-First Approach

**Comparison between POCA vertex and MC Truth**

- Simulation: $\bar{p}p$ elastic scattering at $p_{beam} = 4.06\ \mathrm{GeV}/c$, DPM generator.

| **Parameter** | **MC Truth** | **Prop to $\vec{v}_{\mathrm{MC}}$** | **Prop to $\vec{v}_{\mathrm{POCA}}$** |
|---|---|---|---|
| Vertex $x$ [cm] | $-0.200$ | $-0.197$ | $-0.198$ |
| Vertex $y$ [cm] | $0.401$ | $0.397$ | $0.399$ |
| Vertex $z$ [cm] | $-5.000$ | $-5.001$ | $-5.001$ |
| **Efficiency** | – | $74.2\%$ | $74.4\%$ |

# Performance of POCA-First Approach

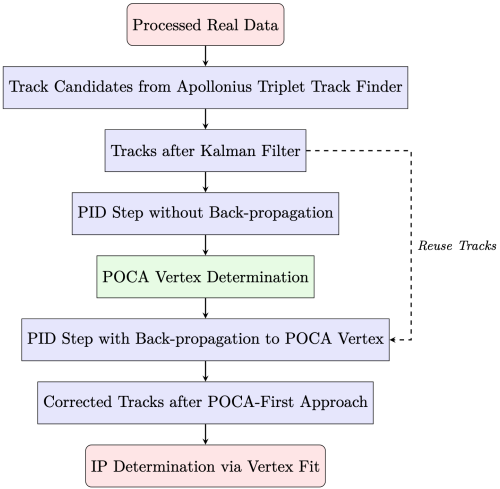
### Comparison with POCA-to-axis

- Simulation: $\bar{p}p$ elastic scattering at $p_{beam} = 4.06\ \mathrm{GeV}/c$, DPM generator.

| **Parameters** | **MC Truth** | **POCA-to-axis** | **POCA-First** |
|---|---|---|---|
| Vertex $x$ [cm] | 0.0 | $-0.0007$ | $-0.0006$ |
| Vertex $y$ [cm] | 0.0 | $-0.0001$ | 0.0002 |
| Vertex $z$ [cm] | 0.0 | $-0.0038$ | $-0.0006$ |
| **Efficiency** | – | 84.4% | 82.1% |

# Workflow for Restgas Determination

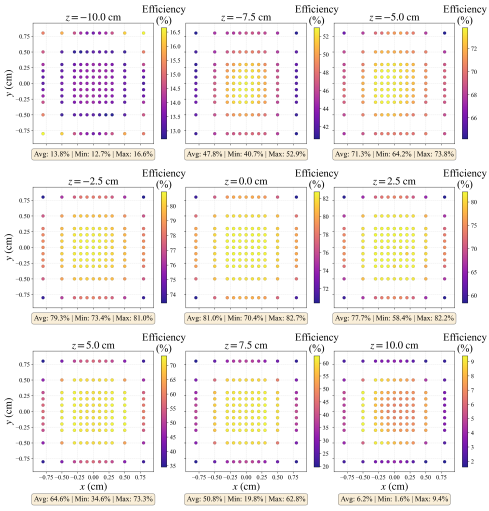
- Determine IP.
- Apply efficiency correction.



# Efficiency Dependence on Vertex Position

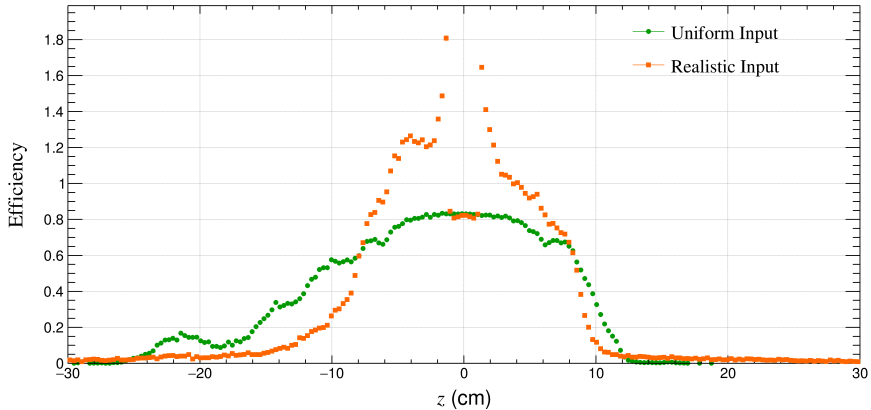
- $p_{beam} = 4.06\ \mathrm{GeV}/c$
- Stable within $|x|, |y| < 0.5\ \mathrm{cm}$.

1D longitudinal efficiency correction is sufficient.



# Resolution Effect

## The "Spill-Over" Phenomenon



# Resolution Effect

**The Forward Modeling Approach**

- Generate MC samples using a realistic input density profile $\rho_{in}(z)$.
- Calculate efficiency $\epsilon = N_{reco}/N_{gen}$

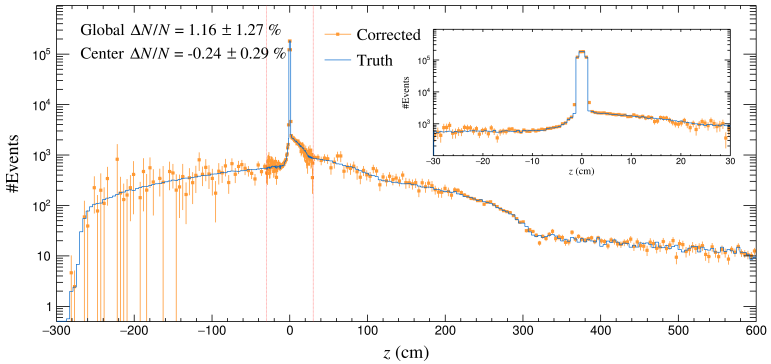
**$\epsilon = N_{reco}/N_{gen}$ acting as a convolved correction function**

effectively encapsulates both:

- Geometrical acceptance.
- Resolution-induced migration.

# Baseline Validation

Input distribution for efficiency perfectly matches the data distribution.




**The Circular Dependency**

Restgas distribution needed for Restgas measurement.

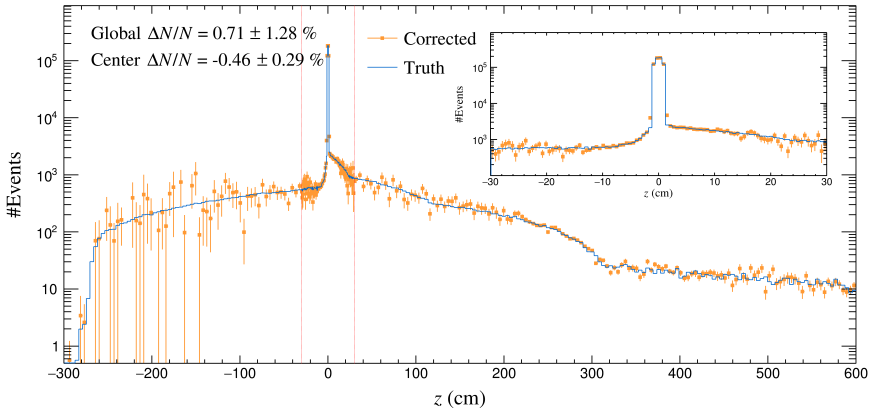
# Convergence Verification

**Stress Tests:**

- A matrix of scenarios are generated:
  - Density of target region kept constant.
  - Density of Restgas background: $0\%, \pm 5\%, \pm 10\%, \pm 20\%$ deviations.
- Each scenario acts as the data distribution and the distribution for efficiency.
- Relative deviation between data and model ranges from $0\%$ to $\pm\, 40\%$.

# Convergence Verification

An Example of $10\%$ Underestimation: *Input:* $-5\%$ vs. *True:* $+5\%$.



# Iterative Workflow

- Analyzed over 30 mismatched scenarios.
- The magnitude of deviation is significantly smaller than the initial input bias.

**The Iterative Results:**

- Difference $< 10\%$ $\Rightarrow$ deviation in center $< 1\%$
  $\&$ global deviation $<$ its uncertainty
- Difference $> 10\%$ $\Rightarrow$ deviation significantly suppressed
  (most of the time in center $< 5\%$)

$\Rightarrow$ The iterative procedure.

# Iterative Workflow

1. **Initialization:**
   Generate efficiency map ($\epsilon$) using Restgas distribution from Target Group.
2. **Reconstruction:**
   Reconstruct IP distribution ($\rho_{reco}$) using ($\epsilon$).
3. **Convergence Check:**
   If central deviation $< 1\%$
   $\&$ global deviation $<$ its uncertainty.
4. **Iteration:**
   If YES, stop and take $\rho_{reco}$ as the determined Restgas distribution.
   If NOT, generate new $\epsilon$ using $\rho_{reco}$ and repeat the loop.

# Iteration Results

An example of iteration starting with $40\%$ overestimation of the Restgas distribution.

| Iteration Step | Global Difference[%] | Central Difference[%] |
|---|---|---|
| 1 | $12.27 \pm 5.69$ | $1.58 \pm 0.30$ |
| 2 | $4.28 \pm 1.19$ | $1.18 \pm 0.28$ |
| 3 | $1.30 \pm 1.32$ | $-0.22 \pm 0.29$ |

The iterative procedure converges rapidly!

# Luminosity Determination

- $p_{beam} = 4.06$ GeV/$c$.
- IP cut around reconstructed IP: $|x|, |y| < 1$ cm, $|z| < 5$ cm

| Target Profiles | Fit Model | $\Delta\mathcal{L}/\mathcal{L}$ |
|---|---|---|
| Point-like Target | Point-like Target | $-0.04 \pm 0.07$ |
| Target with Restgas | MC Target Distribution | $0.02 \pm 0.08$ |
| Target with Restgas | Determined Target Distribution | $0.05 \pm 0.07$ |

# Summary

**A Complete Workflow for Restgas Measurement established:**

- **Reconstruction of Displaced Vertices:**
  The Apollonius triplet track finder combined with the POCA-First approach.
- **Efficiency Correction**
  A forward modeling approach with a rapidly converged iterative workflow.

Luminosity can be determined as precisely as with point-like target or with well-known Restgas distribution to be $\Delta\mathcal{L}/\mathcal{L} = 0.05 \pm 0.07$.



# Thanks for your attention!